

Offline Document Auditor Pro

AI-Powered Contract Risk Analyzer — 100% Offline, 100% Private

Upload a contract (PDF or image), get a risk score, a plain-English summary, flagged clauses with fixes, and a set of tools to actually act on what's found — all without the document ever leaving your machine.

What's New

This update focused on making the tool something you actually negotiate with, not just a one-shot scanner — plus a pass of reliability fixes and a visual refresh.

Feature	What it does
 Clause Library benchmarks	Every flagged clause now shows a "market standard" reference (e.g. typical refund windows, liability cap norms) instead of a generic "negotiate this" note.
 Annotated PDF export	Highlights flagged clauses <i>directly on your original PDF</i> — color-coded by severity, with a margin note (why it matters + suggested fix) — instead of only a separate summary report.
 Negotiation email drafts	One click turns your flagged risks into a ready-to-send email asking the other party to revise specific clauses. Pick a tone (Professional / Firm / Friendly), edit inline, download as <code>.txt</code> .

 Version diffing	Re-upload an edited version of a contract you've already analyzed and see an actual line-by-line diff of what changed, plus the previous risk score for comparison.
 "Explain further" on demand	Click into any flagged clause for a deeper, plain-English explanation (what it does, a realistic scenario where it hurts you, a concrete negotiation tip) — generated only when you ask, so scanning stays fast.
 Smoother loading animations	Fast Mode, Thorough Mode, and Deep AI Analysis now show an animated spinner card with shimmer/fade transitions and a settling "complete" state, instead of a flat static box.
 Reliability fixes	Fixed: annotation highlights obscuring text on the annotated PDF; the flagged-clause list silently capping at 10 even when more were found; and a bug where clicking anything inside Deep Analysis or Thorough Mode results (explain, email draft, downloads) would cause the whole results section to disappear.

Why This Exists

Every day, people sign contracts they don't understand — NDAs, rental agreements, job offers, SaaS terms. Legal review costs real money, and most "AI legal tools" upload your sensitive documents to the cloud. This tool is:

- **Free** — no subscriptions, no API keys
 - **Private** — documents never leave your machine
 - **Fast** — scans in seconds, deep AI analysis in minutes
 - **Actionable** — doesn't just flag problems, helps you fix them
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Features

Core Analysis

- **Fast Mode** — instant regex-based scan (20 risk patterns) across 8 categories: Payment, Termination, Liability, Rights, Data Rights, Legal Rights, Fairness, Control
- **Thorough Mode** — full section-by-section AI analysis using a local LLM, for nuanced language the rule engine can't catch
- **Risk Score** — 0–100 weighted score with color-coded severity breakdown
- **Plain English Summary** — non-lawyer explanation of what's actually risky
- **Clause Library benchmarks** — market-standard reference points attached to every flagged category

Taking Action

- **Deep AI Analysis** (Fast Mode) — one-click LLM pass over the flagged clauses for detailed why/fix reasoning
- **"Explain further"** — on-demand, deeper plain-English breakdown of any single clause
- **Negotiation email drafts** — auto-generated, editable email raising your top concerns with the other party
- **Redlined text export** — suggested edits with strikethroughs
- **Annotated PDF export** — highlights and margin notes written directly onto your original PDF
- **Standard PDF report** — a clean, standalone summary report for sharing

Multi-Document Tools

- **Compare Documents** — upload exactly 2 documents to see side-by-side risk scores and which is safer
- **Batch Processing** — upload 3+ documents to get a ranked list

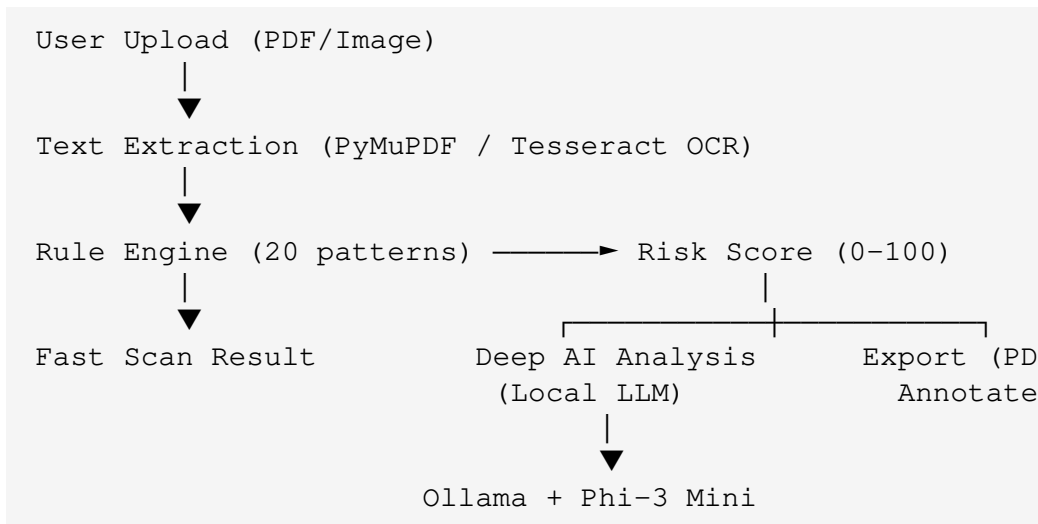
from safest to riskiest

- **Version diffing** — automatically detects when you've re-uploaded an edited version of something already analyzed, and shows exactly what changed

Trust & Privacy

- **Document Fingerprint** — SHA-256 hash for tamper-proofing
- **History Tracking** — local SQLite database of past analyses, per logged-in user
- **No Cloud Uploads, No API Keys, No Telemetry** — everything runs on `localhost`

Architecture



Data flow: Upload → Extract text → Rule-engine scan → Weighted risk score → Optional deep AI pass (local LLM) → Export in whichever format you need.

Tech Stack

Layer	Technology	Purpose
Frontend	Streamlit + custom CSS	Interactive UI
LLM Engine	Ollama + Phi-3 Mini (3.8B)	Local AI inference, no cloud
Text Extraction	PyMuPDF + Tesseract OCR	PDF parsing and image-to-text
PDF Annotation	PyMuPDF	Writing highlights/notes back onto the original PDF
AI Framework	LangChain + LangChain-Ollama	Prompt chaining and templating
Database	SQLite	Per-user analysis history
PDF Generation	fpdf2	Standalone report export
Concurrency	ThreadPoolExecutor	Parallel chunk processing (Thorough Mode)

Installation

Prerequisites

- Python 3.10+
- [Ollama](#) installed
- Tesseract OCR installed ([Windows installer](#))

Step 1: Get the project files

Place `app.py` and `requirements.txt` in a project folder, e.g. `offline-doc-auditor/`.

Step 2: Open a terminal in that folder

```
cd path/to/offline-doc-auditor
```

(On Windows, remember it's `C:\Users\<you>\...`, not `C:\User\...`)

Step 3: Create a virtual environment

```
python -m venv venv

# Windows
venv\Scripts\activate

# macOS/Linux
source venv/bin/activate
```

Step 4: Install dependencies

```
pip install -r requirements.txt
```

Step 5: Download the local LLM

```
ollama pull phi3:mini
```

Step 6: Configure Tesseract (Windows only, if needed)

The app auto-detects Tesseract in the default Windows install location. If yours is installed elsewhere, edit line 9 in `app.py`:

```
pytesseract.pytesseract.tesseract_cmd = r'C:\Program File
```

Step 7: Start Ollama

In a separate terminal:

```
ollama serve
```

Step 8: Run the app

```
streamlit run app.py
```

The app opens at `http://localhost:8501`.

Usage

Create an account

The app requires a local login (username + password, stored hashed in SQLite) so each person's history stays separate on a shared machine.

Single Document Analysis

1. Upload a PDF or image contract
2. Choose **Fast Mode** (seconds) or **Thorough Mode** (a couple minutes)
3. View the risk score, category breakdown, and plain-English summary
4. Click **Deep AI Analysis** (Fast Mode) for detailed per-clause findings

5. Click **Explain further** on any clause for a deeper plain-English breakdown
6. Download: PDF Report, Redlined Text, Annotated PDF, or a Negotiation Email draft

Compare Two Documents

Upload exactly 2 documents and click **Compare Documents** to see side-by-side scores and which one is safer.

Batch Processing

Upload 3+ documents to get a ranked list from safest to riskiest.

Re-analyzing an Edited Version

If you upload a document that's a close match to something already in your history, the app flags it and shows a **"See exactly what changed"** diff view — useful for checking whether a counterparty actually fixed what you flagged in a previous round.

View / Clear History

Use the sidebar:  **View History** to browse past analyses,  **Clear History** to wipe them.

Performance

Metric	Value
Fast Mode	3–10 seconds
Deep AI Analysis	~30–60 seconds

Thorough Mode	2–5 minutes
RAM Usage	4–6 GB (with Phi-3 Mini)
GPU	Not required
Internet	Not required after setup

Limitations

Limitation	Notes
Phi-3 Mini is weaker than GPT-4-class models	Use Thorough Mode for complex contracts; treat flagged clauses as a starting point, not a verdict
OCR accuracy varies with image quality	Prefer native PDFs over scanned images when possible
Annotated PDF highlighting depends on exact text match	LLM-paraphrased quotes sometimes won't be found in the source PDF; the app reports how many of the flagged clauses it actually located and highlighted
Clause library benchmarks are general, not jurisdiction-specific	Treat them as a starting point for negotiation, not legal advice
No legal advice	This tool flags risks and gives a starting point for negotiation — consult a lawyer for anything that actually matters
English only	Multi-language support not yet implemented

Privacy & Security

Feature	Implementation
No Cloud Uploads	All processing happens on <code>localhost</code>
Document Fingerprint	SHA-256 hash for integrity verification
No API Keys	Zero external service dependencies
Local Database	SQLite stored on your machine only
No Telemetry	No usage data collected

Note: contract text is currently stored in plaintext in the local SQLite database (`document_history.db`) to support history and version-diffing. This is fine on a personal machine but worth knowing if the machine is shared or backed up somewhere you don't control.

License

Distributed under the MIT License. See `LICENSE` for details.

Acknowledgments

- [Ollama](#) for making local LLMs accessible
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